A Factorization-Free Certificate for Decimal Mountain Concatenations at Multiplicative Orders at Most 744

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Abstract

Let $M(k) = 1 \parallel 2 \parallel \cdots \parallel k \parallel (k-1) \parallel \cdots \parallel 1$. We certify that every prime $p \neq 2, 5$ with $\mathrm{ord}_p(10) \leq 744$ divides at least one $M(k)$. Our finite computation factors neither $10^r - 1$ nor the large cyclotomic norms appearing in previous research stages. Instead, we combine two elementary infinite-family arguments with exact affine-state computations modulo $10^r - 1$, covering 77 remaining multiplicative orders. The program checks complete small integer gcd obstructions, including a nonprimitive-order contaminant at $r = 696$. This is a bounded-order result, not a solution for all primes.

1 Digit-layer dynamics

For $k \geq 0$, let $C_k = 1 \parallel 2 \parallel \cdots \parallel k$, $R_k = k \parallel (k-1) \parallel \cdots \parallel 1$, and let $D(k)$ be the total decimal digit count from 1 through k. For a modulus n coprime to 10 put

$$S_d = (C_{10^d-1}, Y_{10^d-1}), \qquad Y_k = 10^{-D(k)} R_k \pmod{n}.$$

Within a full d-digit layer define

$$L = 10^{d-1}, \quad q = 10^d, \quad m = 9L, \quad A = q^m, \quad B = q - 1, \quad G = \sum_{i=0}^{m-1} q^i, \quad J = \sum_{i=0}^{m-1} i q^i.$$

The state transition, valid over any such residue ring, is

$$(C, Y) \longmapsto \big(AC + BG - J,\ A^{-1}(Y + LG + J)\big). \tag{1}$$

Only the invertible quantity A is divided by: the expression does not presume $q - 1$ is a unit.

For a prime $p \geq 7$, $p \neq 11$, set

$$r = \mathrm{ord}_p(10) = u\, v, \quad u = 2^a 5^c, \quad (v, 10) = 1, \quad T = \mathrm{lcm}(r, \mathrm{ord}_v(10)), \quad \mathrm{ord}_1(10) = 1.$$

Earlier affine-monodromy lemmas imply

$$S_{jT} = S_T + (j-1)(U, V), \quad (U, V) = S_{2T} - S_T \quad (j \geq 1).$$

The normalized mountain residues at 10^{jT} and $10^{jT} + 1$ are respectively $10C_{jT} + Y_{jT} + 1$ and $1000C_{jT} + Y_{jT} + 121$. If $(U, V) \neq (0, 0)$, their slopes $10U + V$ and $1000U + V$ cannot both vanish because $p \nmid 990$. A linear congruence yields $1 \leq j \leq p$ and a divisible mountain at one of the two offsets.

2 A factorization-free exact-order obstruction

Fix any integer $r \geq 2$, and set $N_r = 10^r - 1$. The residue class of 10 has exact order r in $\mathbb{Z}/N_r\mathbb{Z}$, since $0 < 10^h - 1 < N_r$ for $1 \leq h < r$. Compute the states in (1) in that ring and put

$$(U_r, V_r) = S_{2T} - S_T, \qquad H_r = \gcd(N_r, U_r, V_r). \tag{2}$$

For every prime p of exact order r, reduction of these integer-ring states modulo p gives the prime-field drift. Consequently

$$(U, V) \equiv (0, 0) \pmod{p} \quad \Longrightarrow \quad p \mid H_r. \tag{3}$$

Computing (2) does not require building the huge $m = 9 \cdot 10^{d-1}$. Put $h = r/\gcd(r, d)$ and write $m = th + b$, $0 \le b < h$. Since $q^h = 1$,

$$G = tG_h + G_b, \qquad J = tJ_h + hG_h\binom{t}{2} + thG_b + J_b.$$

Use modular powering to find $m \bmod hN_r$, hence $t \bmod N_r$ and b. Binary splitting evaluates G_h, J_h, G_b, J_b with $O(\log r)$ residue-ring operations. As N_r is odd, $\binom{t}{2} \bmod N_r$ is determined by $t \bmod N_r$. Thus the entire check uses exact arithmetic without factoring the repunit N_r.

Theorem 1 (Certified bounded-order existence)**.** *For every prime $p \neq 2, 5$ with $\mathrm{ord}_p(10) \le 744$, some positive k satisfies $p \mid M(k)$. For $p \ge 7$, $p \neq 11$ we may take $k = 10^{jT}$ or $10^{jT} + 1$ with $1 \le j \le p$.*

Proof. The cases $p = 3$ and $p = 11$ follow respectively from $3 \mid M(3) = 12321$ and $11 \mid M(2) = 121$. For every other p, partition its order $r \le 744$ according to $u = 2^{v_2(r)}5^{v_5(r)}$ and $v = r/u$.

If $u \mid 20$, the previously proved elementary finite-defect transfer gives a perfect-square first-window polynomial with a subgroup root. If $u \nmid 20$ and $v \mid 9$, the elementary invariant

$$U + V = 9 \cdot 100^{-1} \pmod{p} \neq 0$$

gives nonzero drift.

Exactly 77 order values $2 \le r \le 744$ lie in neither elementary family. For all 77, the included programs compute H_r in (2) and completely factor this *small* integer by trial division. They check $\mathrm{ord}_q(10) \neq r$ for every prime factor $q \mid H_r$. Therefore no prime p of exact order r can satisfy (3), and its drift must be nonzero. The two-offset lemma gives the witness. □

3 Certificate, audit, and limitations

The 77 exceptional-to-the-elementary-family orders consist of 42 previously checked order classes $r \le 500$, 31 new classes $501 \le r \le 744$, and four former high-3-power cases:

r	216	432	648	675
H_r	243	243	729	243

Some representative new values are

r	504	536	584	624	664	696	725	744
H_r	567	99	657	351	9	189	9	27

At $r = 696$, $H_r = 189 = 3^3 \cdot 7$; the extra factor 7 is harmless because $\mathrm{ord}_7(10) = 6 \neq 696$. The program therefore tests exact orders of all gcd prime divisors, rather than assuming the false stronger assertion that every gcd factor divides $990r$.

The 35 additional exact (r, T, H_r) rows are pinned with SHA-256

`211b961ab1c7c1762aa39df1e16415a0e56296ef5b4a1a9a9ffbec3d305d5aa3`.

An independent prime-field recurrence verifies

r	p	$U_r \bmod p$	$V_r \bmod p$
584	1753	1438	1332
664	1993	1889	1956
696	13921	13258	9689

The exact, standard-library Python programs are `certify_orders_le500.py`, `certify_orders_le744.py`, and `audit_order_744_prime_field.py`, in the accompanying `research/` directory.

The bounded-order proof requires independent auditing of the two previously obtained elementary lemmas and of the short exact-arithmetic programs. It neither proves universal nonzero drift nor resolves OEIS A077187 for all primes.